

New method and MATSim contrib for the Agent-based Traffic Assignment Problem (ATAP)

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Introduction

In MATSim, travel demand is represented by integral decision makers, travel plans may include route, departure-time, and mode choices, and network loading is realized through a vehicle-discrete simulation. This modeling approach offers substantial descriptive flexibility compared to classical continuum-flow assignment, but it also makes it more difficult to compute travel plans that are consistent with the traffic conditions that they create.

The standard MATSim approach relies on co-evolutionary replanning, where selected travelers modify their plans and keep successful alternatives. While this approach works well in practice, it is not explicitly formulated as a traffic assignment procedure. The Agent-based Traffic Assignment Problem (ATAP) contrib was developed to provide a more assignment-oriented framework for MATSim while preserving the behavioral richness and simulation-based nature of agent-based models (Flötteröd, 2026a, 2026b).

Method

The approach starts from a formal representation of the traveler- and vehicle-discrete assignment problem using a Nikaido–Isoda (NI) gap function. The NI gap quantifies how much travelers could improve their utility by switching from their currently selected plans to alternative plans. A challenge is that direct optimization of the NI gap is not operational in realistic large-scale simulations. The method therefore derives an upper bound on the NI gap and reformulates this bound as a replanner-selection problem. The resulting assignment logic balances two objectives: maximizing anticipated utility improvements and minimizing anticipated changes in network conditions. This leads to an endogenous replanner-selection mechanism. Instead of specifying in advance how many travelers may change plans in a given iteration, the method determines this number from the assignment problem itself. A single ambition-level parameter controls how aggressively the algorithm seeks gap reductions.

Stochasticity and vehicle discreteness are addressed through utility averaging across assignment iterations and a flow-based approximation of network conditions based on temporally smoothed link-entry flows. The flow-based approximation is also used to anticipate changes in network conditions. ATAP further anticipates the scores of not-yet-executed plans by executing these plans outside the main MATSim simulation using fixed travel times. This functionality is similar to PSim (Fourie, 2016; Fourie, 2026).

The method remains largely agnostic to the internal structure of both the behavioral and network loading models. It requires only candidate plans, utility evaluations, and a representation of resulting network conditions.

Practical results

The ATAP contrib supports different replanner-selection strategies, including uniform switching, utility-based sorting (Sbayti et al., 2007), and the new NI-gap-based assignment heuristic. In practical terms, the assignment seeks solutions in which individual travelers can improve their utility as little as possible through unilateral plan changes while also reducing assignment noise and unnecessary fluctuations in network conditions across iterations.

The NI-gap-based assignment method has been evaluated in large-scale MATSim applications for Stockholm, Oslo, and Berlin under varying congestion levels and behavioral settings. The experiments compare the proposed strategy against both uniform replanner selection and utility-based sorting approaches. Across the considered case studies, the proposed method consistently achieves lower assignment gaps. The experiments indicate that explicitly accounting for anticipated changes in network conditions provides a practical way to stabilize assignment dynamics. This becomes increasingly important as congestion levels rise and traveler decisions have a stronger influence on subsequent network conditions.

These results show that formal assignment concepts can be transferred to simulation-based transport models without sacrificing behavioral richness or requiring strong modeling assumptions.

References

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